Exp:6
4/9/26

ERROR DETECTION AND CORRECTION USING HAMMING CODE

Aim:

To write a pgrm to implement error detection and correction using Hamming Code concept

Make a test run to input data stream and verify error correction feature.

SENDER:

* Input to sender file should be a text of any length program should be convert the text to binary.
* Apply hamming code concept on the binary data and add redundant bits to it.
* Save this output in a file called Channel.

CODE:

```
data = input("Enter 4 bits:")
d1, d2, d3, d4 = map(int, data)

    p1 = d1^d2^d4
    p2 = d1^d3^d4
    p3 = d2^d3^d4

code = [p1, p2, p3, p4, d2, d3, d4]
print("sender hamming code:", join(
        map(str, code)))
```

RECEIVER SIDE:

* Receive the hamming code from the sender.
* Enter the recived code.
* calculate the parity checks.
* Find the error position using possible values.

CODE:

```
r=list(map(int,input("Enter recived bits:")))
r1 = r[0]^r[2]^r[4]^r[6]
r2 = r[1]^r[2]^r[5]^[6]r
r3 = r[3]^r[4]^r[5]^r[6]
error = r1 * r2*1 *2 * r4 * 4.
if error==0:
        print("NO error")
else:
    print["Error at position, r[error-]^=1
    print("corrected code:", ".join(map,
                                    recived)))
```

Result:

Error detection and error correction using Hamming Code was executed successfully.